

WASP-10b

R-1154

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_201001 · created 2026-07-31T01:33:53+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459123.7443 +/- 0.0046 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.043
Transit depth (Rp/Rs)^2	2.72 +/- 1.41 [%]
Orbital Inclination (inc)	89.47 +/- 1.41
Airmass coefficient 1 (a1)	1.285 +/- 0.022
Airmass coefficient 2 (a2)	-0.24 +/- 0.29
Scatter in the residuals of the lightcurve fit is	1.7 %
Best Comparison Star	#2 - [513, 307]
Optimal Aperture	2.89
Optimal Annulus	13.71
Transit Duration (day)	0.0342 +/- 0.0059

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.165 $\pm$ 0.043 vs 0.1592 (deviation 0.13 σ)
Duration fit vs published	0.0342 d vs 0.0946 d (deviation 10.23 σ)
Mid-time O-C	-10.9 min
Depth SNR	3.8
Residual scatter	1.7 %

Light curve

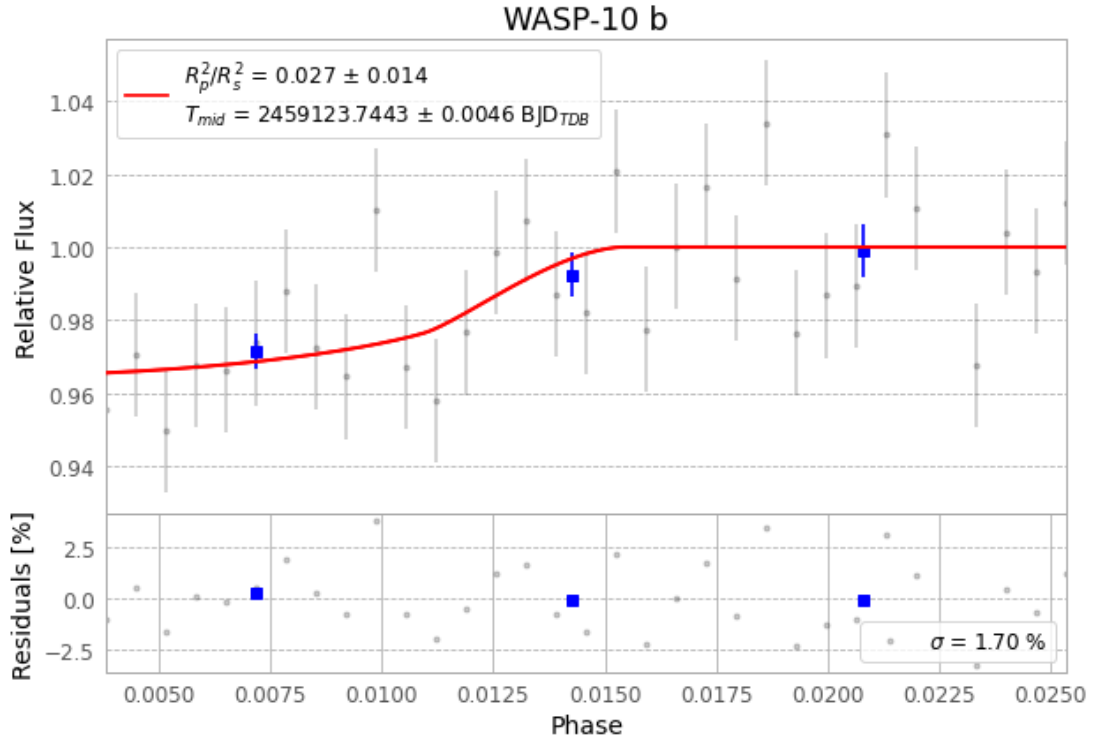


Figure 1 — FinalLightCurve_WASP-10 b_2020-09-30.png (SHA-256 cfc23ea8f6d7d389...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [285, 297], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c2f399ef7d6c47d0...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

33 FITS frames (22 MB), WASP-10201001060014.FITS → WASP-10201001073613.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-201001.log (321aba17fb0c...), FinalLightCurve_WASP-10 b_2020-09-30.png (cfc23ea8f6d7...), FinalLightCurve_WASP-10 b_2020-09-30.csv (05ce2b00539e...)

Compute resources

Wall time 215.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 01:33Z.